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***published in***
Research Handbook on Nudges and Society
2023

***DOI (link to publisher)***
10.4337/9781035303038.00023

***document version***
Publisher's PDF, also known as Version of record

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***citation for published version (APA)***
Banerjee, S., & John, P. (2023). Nudge+: Putting citizens at the heart of behavioural public policy. In C. R. Sunstein, & L. A. Reisch (Eds.), *Research Handbook on Nudges and Society* (pp. 227-241). Edward Elgar Publishing Ltd.. https://doi.org/10.4337/9781035303038.00023

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# 13. Nudge+: putting citizens at the heart of behavioural public policy

*Sanchayan Banerjee and Peter John*

## INTRODUCTION

The success of behavioural public policies (BPPs), occurring over the last 15 years or so, presents scholars and policymakers with an opportunity to scale up the knowledge that has been gained (Al-Ubaydli et al., 2021; Banerjee & Mitra, 2023). Nudges – changes in how choices are presented to citizens – could encompass more policy domains and contexts with more robust and effective interventions. To complement this scientific advance, we argue that the democratic legitimacy of nudges could also be fostered. Essential to the design and delivery of public policies in contemporary democracies, citizens need to engage more with BPPs rather than act largely passively as they do at present. So, how can BPPs be made more citizen-friendly? There are various ways in which this aim could be realised (Banerjee et al., forthcoming), some of which appear in other parts of this handbook. Our approach is to introduce a thinking element into BPP whereby reflection becomes part and parcel of the design and delivery of nudges. We call this modified BPP toolkit, Nudge+ (Banerjee & John, 2021).

By enabling citizens to reflect on nudges, Nudge+ creates a two-way channel of communication between citizens and the state, whereby each can feed back to the other on the design and implementation of public policies. Nudge+ ensures that citizens engage with public policies yet, at the same time, they and society benefit from the deployment of the tools of behaviour change. Nudge+, then, harnesses the best from nudge and think deliberations.

This chapter is a review of the Nudge+ tool, first explaining why a deliberative or thinking element is needed for BPP at the present time, locating where Nudge+ comes from intellectually in the history of think and deliberative interventions, and summarising the argument in favour of a modified nudge rather than full-on deliberative engagement with citizens. Then, it provides a definition of Nudge+ and asks what happens when people reflect, which is followed by a review of the extant research evidence as well as an assessment of the possible limitations of Nudge+ and how they may be overcome.

## BACKGROUND AND ORIGINS

The advantage of the traditional economic or rational-actor model of decision-making is that no matter how far it is from the reality of how day-to-day thought processing



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works, it puts individual choice at the centre of public policies. Individuals may be subject to decisions made a long way from people's locales and peer groups, and are objects of policies rather than drivers of them; but at least they are assumed to make choices based on conscious calculations of ways that get them to reach their objectives, if in the context of constraints imposed by policymakers. Even criminals, when deciding whether to commit crimes, are assumed to be making calculations against costs and benefits, with policymakers putting in place costed disincentives to ensure society is protected (Becker, 2000). Such calculations could not happen by heuristics and short-cuts alone (even though these can help sometimes) but from self-awareness of objectives (e.g. to make money from crime), knowledge about the costs of taking action (probability of being caught, obstacles to entering the property or other barriers in the way of seizing other people's resources), and a calculation of the benefits received (inflow of funds that are not detected by the police, elan at outwitting the police). In the BPP world, choosing individuals are replaced by those who respond through their intuitions, biases, and heuristics. Policymakers can tailor interventions and redesign choice architectures to take advantage of these biases and ensure individuals take desired public actions (Thaler & Sunstein, 2009).

Much of this influence happens unconsciously, with BPPs thought to work best 'in the dark' (Bovens, 2009). In the nudge worldview, this relative passivity is portrayed as consistent with life choices, going along with people's preferences and involving low cognitive effort, which people are assumed to prefer. Liberty is preserved because if people become aware of nudges they can opt-out of them afterwards, hence libertarian paternalism (Sunstein & Thaler, 2003; Thaler & Sunstein, 2003).

Critics, and even those sympathetic to BPP (e.g. Anderson, 2010; Sugden, 2009), have pointed out that this proviso does not meet the normal conditions of autonomy and agency whereby individuals make active decisions at the point or before the choice is made not afterwards. Moreover, it seems odd that people might only by chance find out about what has been done to them in their name, if they come across the policy or search for it on government websites. This relative covertness is different to many standard policies, such as taxes and regulations, for which there is often a public effort to communicate to citizens, usually as a necessary condition of them working (e.g. publicising an anti-smoking ban). Nudge advocates argue that a lack of practicality and potentially spoiling of the interventions mean that the policymaker can't advertise them too much in spite of respecting the publicity principle. It is rather like researchers who cannot directly tell patients they are in a randomised controlled trial (but for transparency leave details of the trial on their websites that no patient is likely to read). These researchers need to carry out a debrief to explain why consent was not sought at the time the experiment was delivered. As with researchers, nudgers are acting ethically in taking this retrospective stance, but that the ethical quality of BPPs could be improved by allowing for early reflection and greater self-awareness (John, 2022). Individuals need to be allowed to make an active choice, which is in addition to publicity and transparency.

To offer an active role for citizens, BPP needs to move away from its largely top-down welfarist perspective to one where individuals can drive public policy, at least

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to an extent. Of course, traditional top-down public policies can embed individual freedom through the decisions of representatives mandated by elections. In a classic representative democracy, the citizen role is limited to voting retrospectively for an incumbent who can select policies (including nudges) (Fiorina, 1981). Administrations are to a degree responsive between elections so as to maintain electoral advantage (Manin et al., 1999). But voting decisions are not always efficient, especially when tailoring policies for heterogeneous individuals; moreover, vote cycling and other inconsistencies do not aggregate a simple set of preferences (see review by List, 2018). In recent times, democratic values have been seen to be more in play when public administrations are to a degree responsive at the time decisions are made and there is some direct input into that decision-making (Besley & Burgess, 2002; Warren, 2009). Since the 1970s, a powerful movement has grown up to challenge traditional public administration, drawing from the participatory ideas of pluralists (Hirst, 1994) and radicals (Barnes, 1979). In response, public administrators have set up mechanisms to include citizens in decision-making through consultation and setting up decision-making forums, called citizen governance (John, 2009a, 2009b) or empowered participation (Fung, 2004; Fung & Wright, 2003). At its most pronounced, mini-publics and deliberative institutions can take decisions alongside or instead of elected representatives (Smith, 2009), such as through deliberative polls (Fishkin, 2011).

The programme of think draws from these proposals for mini-publics, but seeks a more direct connection to public policy in helping solve implementation and delivery challenges in partnership with citizens. Mini-publics can be turned into thinking-style BPPs or deliberative nudges (Sunstein, 2016). These thinks can be designed in collaboration with policymakers, and were tested alongside nudges in a series of experiments carried out in the 2000s (John, 2011). These experiments show that thinks can engage citizens, but they are hard to scale up and achieve concrete policy solutions, such as an organ donation deliberation treatment arm that got a worse outcome than the placebo. Both deliberations and thinks impose too high a cognitive burden and time cost, leading to the self-selection of subjects into these interventions with small non-representative samples taking part (Smith et al., 2013).

The project of Nudge+, first called Nudge Plus (Banerjee & John, 2021; John, 2018; John & Stoker, 2019), later shortened to Nudge+ (Banerjee & John, 2022), emerges from these findings and debates, aiming to have a more economically and targeted form of think that does not give up on reflection, nor on the power of nudges, by combining both think and nudge. Nudge+ procedures impose less time and cognitive costs on individuals than think, so encouraging greater participation and more representative samples than most mini-publics tend to achieve. Reflection allows people to decide consciously to go along with what the nudge indicates they should do, or alternatively not to take this course of action, making the choice active and not just assumed to be subject to an after-the-event revocation. They can freely accept or reject the offer of the paternalist at the time the nudge is administered.

Nudge+ takes advantage of existing knowledge about improving policy outcomes from the behavioural sciences. Transparency interventions show that telling the person getting nudges does not reduce its effectiveness, as nudges do not need just to rest

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on unconscious processes (Bruns et al., 2018; Loewenstein et al., 2015; Michaelsen et al., 2021; Wachner et al., 2020). There may be a benefit that people own the process and are more willing to engage in behavioural change. We present findings later in this chapter to show that Nudge+ does work better than standard nudges, so banking the benefits of nudge, but adding more impact on individual and collective welfare, potentially for the long-term, hence additional rationale for the plus or +. Moreover, when citizens understand the reasons for their actions, they may be more likely to embrace long-term changes in their behaviours, rather than just reacting automatically to nudges with no rationale to keep going other than habit. Nudge+, then, may be a route to sustained behaviour change.

It has been suggested recently that standard behavioural public policies focus too much on the individual (the i-frame), rather than taking a system, holistic, and institutionally sensitive perspective (the s-frame) (Chater & Loewenstein, 2022), which is echoed in Hallsworth's manifesto for behavioural science (2022) calling to 'see the system'. While nudges should be embedded in the wider system, either to formulate new public policies independently or in association with traditional tools of governance as hybrid policies (Gravert & Shreedhar, 2022; Loewenstein & Chater, 2017; Reisch, 2021), it is incorrect to assert generally that policymakers, researchers, and practitioners overlook this feature in the design of BPPs. The attraction of Nudge+ and other agency-enhancing interventions, such as boosts (capacity-augmenting measures) (Hertwig & Grüne-Yanoff, 2017), is that they bring the individual and structural perspectives together. Thinking-based nudges offer scope for the s-frame because they ask individuals to be more aware of the reasons why nudging is needed and what problems it can address, with greater understanding of the chains of public action and decisions surrounding nudges. Successful public policies, such as nudges, involve collaborations of many actors, the overcoming of obstacles to implementation, clarity and coordination from policymakers, and most of all active collaboration with citizens who need to step up into a policy-formation role rather than being just being a passive recipient of services. The additive feature of the plus or + strengthens the credibility of nudges as it helps individuals think about their role in this complex social system. Nudge+ brings the i- and s-frames together and does not see them as opposed.

Nudge+ can be tailored to deal with collective action dilemmas as well as individual ones. In many cases, encouraging individuals to change their behaviours also helps collective outcomes (e.g. better diets reduce obesity so help collective outcomes such as less spending on publicly funded health services). In other cases, the collective and individual outcomes are not so aligned where the individual will not act unless they see others act, the classic collective action dilemma (Hardin, 2015; Ostrom, 1990). Classic nudges may not address these collective problems, but thinking-based nudges might be able to because they can encourage people to reflect on the behaviour of others, so creating signals that encourage people to cooperate, as in classic game theory (Sandler, 2004). With common resource dilemmas, such as over the environment, people can learn to cooperate to protect collective resources, such as water supply (Ostrom, 1990), and by extension the environment as a whole, such as the climate (Harris, 2007).

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Of course, thinking might mean that people are made more aware of collective action dilemmas and be less likely to act as they get to realise the nature of the payoffs. People become more selfish by letting them think! Defection in the prisoners' dilemma game might increase with reflection. Also, selflessness can lead to lack of cooperation as all the payoffs in the game are reversed (Collard, 1978). Maybe over climate change people think about the impossibility of a successful result given lack of international cooperation, making their efforts at, for example, recycling seemingly pointless. Altruists might think they are better off directing their energies to poverty, for example. Welfare-orientated policymakers might be forgiven for not wanting to be too transparent. But reflection can lead to more altruism (Niker, 2018). Moreover, collective action problems can be overcome through agreement on rules (Ostrom, 1990, 1998), which need thinking and debate to agree.

One important proviso is people do not always change their behaviour in the direction favoured by the policymaker. In fact, given agency, an individual is entitled not to respond in the required direction favoured by the paternalist, but to say the required behaviour is not for me. As a result, Nudge+ interventions can backfire. Rather than being a problem, backfires might be one of the attractions of this bottom-up citizen-focused BPP. This stance need not be just a libertarian respect for individuals who reject welfare-enhancing outcomes, who can say, for example, they prefer to drink, smoke, and eat unhealthy foods. Rather, it is hard for policymakers to know for certain what are the exact interests of individuals, which might vary from person to person (White, 2016). For example, someone who wants to move from country to country might not need to join the company pension scheme, preferring to save in a way that is more portable rather than leaving behind savings in a myriad of complex funds that might be hard to access later in life. Classic nudges, such as a pensions default in this example, entail just one response to the nudge; Nudge+ can envisage several legitimate responses, encouraged by the think. The side-benefit of this open strategy is that it shows that policymakers respect individuals and their circumstances rather than assuming that one size fits all. More trust by government in individual decisions could have advantages of increasing reciprocal trust in government, which has been so shattered in recent years.

## WHAT EXACTLY IS NUDGE+?

From our discussion so far, it should be clear that Nudge+ is a hybrid toolkit of BPP, one that combines nudge and think in a specific bundle. As we have written elsewhere (Banerjee & John, 2021, 2023), this can be done differently but there are two main ways of designing this tool. In its first form, large-scale independent thinking with think is delivered before individuals are nudged, making such a combination a two-part Nudge+. However, since thinks are hard to scale up at the population level, in a Nudge+ set-up, they are replaced by mini-think prompts, which can facilitate individual reflection before a set of nudges are delivered. These interventions, therefore, are two-part because the Nudge+ is usually delivered into two different parts, the mini-think, and the nudge, both designed for the individual citizen. These

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interventions have the additional requirement that they must be carefully made proximate to each other, such that neither of them resembles a full nudge or a think but complement each other in ways that realise the true potential of a Nudge+. The think allows individuals to debias (see Croskerry et al., 2013) themselves of the undue influence of the nudge, whereas the nudge makes it easier and compliant for the individual to follow-up with their reflected decisions following the think.

Let's outline how a two-part Nudge+ intervention works. Consider the nudge, a commitment device. A commitment contract works by keeping people on track for realising certain end goals or targets. However, implicit in this commitment contract are two active decisions which an individual must take before they can fully deliver on the potential of this nudge. First, when a commitment device or a pledge is provided to a person, they must think whether their own goals conform with the goals of the device, conditional on which they sign the contract. Second, if these goals match, then individuals must also think of the best possible way to keep doing the right behaviour in order to comply with their commitment made through the device. Every commitment nudge has some passive element of reflection embedded into them, which can be turned into an active decision mechanism – simply by making these two steps explicit to people. When done so, a passive commitment nudge becomes an active commitment think. However, this think is hard to comply with, simply because it involves a series of thoughtful decisions, the latter requiring a series of repeated good decisions. To reduce the burden of this think, it is possible to upgrade the commitment nudge into a Nudge+. This process works as follows: encouraging people to think as the first step to find out whether their goals match that of the pledge and to generate an explicit signal from citizens for policymakers. In turn, policymakers can relieve citizens of the cognitive burden of thinking in the second stage by nudging them into ways that keep them compliant with their initial commitment. Thus, in the second stage, policymakers can nudge committed individuals into the desired behaviour, say with an automatic default which endows citizens with a choice which they can continue with. This way of eliciting individuals' thoughtful preferences, without passively assuming their commitment or without automatically defaulting them into a certain option, is ethically superior as it empowers people to form their opinions without being unduly influenced by the nudge in the first instance. In cases when citizens do not reveal preferences that match the goals of the pledge, policymakers should be deterred from mindlessly nudging citizens into that direction – it is here that other kinds of public policies, but not nudges, are best suited.

It is important that sequencing of the nudge and the think in this combination can be easily reversed, based on the context of making these decisions. For example, it is possible in multi-shot interactions that individuals are nudged first, before being asked to reveal their commitment, and they may revisit the nudge. Of course, while such sequence reversals are plausible, their rationale and desired effectiveness are context-, situation-, and outcome-dependent.

In contrast to these two-part tools, another kind of Nudge+ is a one-part device, whereby the nudge and the think are carefully pieced together and embedded into one tool, such that it is not possible to separate them out into different components. Hence, their name, one-part. These one-part Nudge+ interventions are designed to prompt

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reflection in citizens while simultaneously nudging them towards certain ends. Let's revisit the example of the commitment device to make this point. However, it is possible over time, as citizens work towards realising those ends, they update their beliefs and preferences. In such cases, a standalone nudge, like the one above, which lacks an element of reflection or citizen engagement in it, becomes welfare-reducing for citizens, as it does not enable people to revisit their original commitment and update it over time. Here, there is potential to upgrade the nudge by building multiple feedback loops between the citizen and the nudge, whereby the citizen can reflect on the nudge at different stages of its operation. Consider, for example, the modified dual-self-commitment device. This is based on the psychological idea that all individuals have two different versions of their own selves: a short-run self and a long-run self, and sometimes these two versions become inconsistent with one another. As such, the dual-self device elicits two different commitment targets for citizens, a short- and a long-run target. Every time individuals complete a short-run period, the nudge then feeds back to citizens about their performance, particularly in reference to their long-term targets, and citizens are encouraged to think and revisit their long-run target, as they are set on to their next short-run target. In this way, a dual-self device is an example of a one-part nudge which not only nudges people towards certain end outcomes but also enables people to reflect on their journey, achieve their own commitment, and revisit it from time to time. These dual-self-commitment devices can be applied in many different settings, such as when people sign up to a gym and set their long-term weight targets, or when customers set up their digital purchase or minimum spend limits.

It is important that this reflection in the dual-self-commitment device is sequentially delivered to the receivers of the Nudge+ tool. In other words, users of the dual-self device are encouraged to think only after a certain period has passed, which in turn makes this combination sequential in design. However, it is also possible to develop a one-part Nudge+ tool which simultaneously delivers the thinking prompt with the nudge. As pointed out earlier in Banerjee and John (2021, 2023), an example of this is a smart device, like the GPS which nudges people for navigability, combined with AI prompts to encourage reflection on the best use of the device. Similarly, other forms of smart devices, like smart meters, which nudge people into making certain energy decisions, combined with AI prompts for human engagement and reflection are also examples of the one-part Nudge+.

As this example shows, Nudge+ relies on a particular take on cognition as a hybrid between the fast (System 1) and the slow (System 2) means of thinking. It is still axiomatic that there are two systems at play in the brain, and that nudge interventions aim to bypass actions originating from System 1 (Kahneman, 2011). Think by implication uses System 2, slow thinking. As Banerjee and John (2021) review, in practice the brain works often with both Systems 1 and 2 in play at the same time, not far from each other, or ready to switch between the two if needed. Adding the reflective plus to the nudge facilitates this switch by activating the gatekeeper to watch out for responses under the nudge, just to make sure people conform with their own preferences and goals when under its effect. This is just how Nudge+ works by harnessing the best from System 1 and System 2, just as it gets the best from nudge and think in ways that are convenient for individuals as well as for society.

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## THE ROLE OF PERSPECTIVE TRANSFORMATION IN NUDGE+

A Nudge+ is based on the idea of perspective transformation, which is the core principle underlying any process of reflection. When a prompt of reflection is delivered to a citizen alongside a nudge they receive a certain stimulus, which enables them to evaluate the nudge and then decide on their response to it: either they reject the stimulus, because the nudge does not conform with their own values, or they accept it, in which case the nudge towards a certain direction is well-received by citizens. Either way, the reflective prompt allows people to become debiased on the undue influence of the nudge and develop their opinions freely. This process restores the agency of the citizens. It allows them to develop freely their own choices and evaluate the nudge on its own merits without being unduly influenced by it towards any direction.

Let's consider an example. Suppose a policymaker wants to steer citizens into eating plant-based diets only. They decide to do this by changing all menus in any restaurant to a smaller green-menu which displays plant-based items only. This is a nudge since all customers are automatically defaulted to the environmentally-friendly items. Of course, customers can ask for a bigger menu, and while ordering they can also easily opt-out of the plant-based default menu which is provided by the restaurant.

Now consider a reflective prompt for all customers in this restaurant: this prompt could be in the form of an information disclosure that is provided to all incoming customers explaining the reason behind the construction of this default menu. What does this reflective prompt do? For customers, this reflective prompt is like a knee-jerk reaction. When people receive this prompt, they pause for a second and they think 'So, the restaurant wants me to eat green today. Do I want to be nudged in this way?' Based on the response they come up with to the question, they then decide whether they want to follow the nudge or override it. This process in entirety is called perspective transformation as people receive a stimulus, which allows them to develop or update their own perspectives and then devise a response appropriately. As it is clear from this example, adding a reflective prompt to the nudge can lead to two possible outcomes: one where the citizens decide to go along with the nudge (desired effectiveness) or another whether they decide to block the nudge (leads to a backfire). The advantage of a backfire, in this case, is that while the policy fails, it does create some signal between the citizens and the state which conveys the true and revealed preferences of the individuals in this instance. This is how Nudge+ can put citizens into the heart of BPP. It does not mindlessly nudge people in certain directions, but it empowers individuals to judge a nudge on its own merits before deciding whether that is the best means of behaviour change. And adding the information disclosure can be cost-effective too – there could be multiple ways of doing it, for example through flyers distributed alongside these new menus, or signage displayed in the cafeterias, or even through the self-screen interactive checkouts like in popular fast-food joints McDonald's and Kentucky Fried Chicken (KFC).

The two-way relationship between citizens and the state can be extended by putting the initiative in citizen's hands to promote thinks by elites. We know that elites can be the target of interventions designed to reveal their behaviour, such as through audit experiments on politicians to see how much they are prejudiced (Butler, 2014), which have been criticised, as with critiques of nudges, that they do not seek the

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consent of politicians (John & Foos, 2022; Loewen & Rubenson, 2021). Citizens could use the same method of targeting MPs but in a way to get them to think about their behaviour, such as in deliberation experiments done in partnership between citizens and politicians (Neblo et al., 2018).

## EXTANT EVIDENCE ON NUDGE+

Nudge+ is a new research programme, which has been gaining traction over the last few years. It is encouraging to see evidence in favour of Nudge+ studies, both in trials led by the authors themselves and by other scholars in the field. In this section, we summarise the Nudge+ studies that have been conducted over the last few years.

The first Nudge+ trial was conducted by Monica Mühlböck and co-authors between May and December 2017, using a group of 7,000+ randomly selected Austrian unemployed young people. The findings were later reported by Mühlböck et al. (2022) In this trial, the authors designed four different interventions using randomised combinations of an 'information nudge' and a 'reflection nudge' that was designed to increase the probability of a successful job search amongst these unemployed Austrian youths. They showed that when the participants were randomly encouraged to reflect on elements that were needed for a successful job search before nudging them with the required information, it increased the probability of a successful job search within six months by 3.6 percentage points. Based on this, the authors concluded that reflection enabled people to make better use of the information for their job searches.

Following this, we designed a Nudge+ experiment to compare systematically four different behavioural toolkits, namely the nudge, think, boost, and Nudge+. In this online survey experiment, more than 3,000+ people in the United Kingdom, recruited via Prolific, were randomly assigned to ten different experimental conditions, to improve the uptake of sustainable diets. These findings were later reported by Banerjee et al. (2022a). In this experiment, we found that eliciting people's commitment to sustainable diets, before defaulting them into similar diets, increased the uptake of the diets by 30 per cent or more versus nudging them without any reflection prior to it. The results also showed that transparent nudging was as good as nudging itself, thereby adding to the list of findings that informing people about the nudge did not reduce its effectiveness. In this set of experimental conditions, Banerjee et al. (2023) further reported that nudging people who were already motivated to eat sustainably in the short-term, after they had taken the pledge, led to a backfire. This backfiring effect, nonetheless, faded out when people's overall long-term motivations were controlled for – with the policy implication that policymakers should not be deterred by initial reactance to BPP if such effects are driven by people's short-term motivations only.

To check the validity of these findings, Banerjee and Picard (2023) replicated the role of reflection in a nudge with a different nudge: social norms. In another experiment with more than 5,500 participants in the United Kingdom, also recruited by Prolific, they showed that when people are made to reflect on the social norms with their personal norms, and then take a pledge to conform to the said social norms, the effectiveness of the social norms nudge almost doubles compared to those who do

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not reflect on personal norms or take the pledge. This experiment showed the importance of norm internalisation, i.e. the ability of a person to accept norms and relate to them, and successfully driving behaviour change with it. As the authors showed, simply identifying personal norms was not sufficient to improve on the effectiveness of the social norms – only when participants could identify these personal norms first and relate these personal norms with the social norms next, by taking a pledge to conform to the latter, the effectiveness of the norms nudge increased. Recently, Thamer (2023) evaluated a labelling nudge versus a nudge+ (labelling with reflective prompts for individuals) randomly using a field experiment with 227 students in German university cafeterias, and showed that encouraging reflection in participants alongisde the nudge improves the uptake of vegetarian meals.

Later, in a team of social scientists across the United States, United Kingdom, and Canada, we designed and administered a Nudge+ experiment to more than 24,000 participants who were representative of age, gender, and education profiles of their respective populations. In this experiment, later reported by Banerjee et al. (2022), we designed a hypothetical two-by-two nudge-think factorial experiment, where in the nudge dimension, people who received the treatment were told they would be automatically defaulted by their local health clinic into receiving a vaccine at their convenience. Otherwise, they were told about a vaccine readily available for them to take and they should arrange to receive it themselves. Similarly, in the think dimension, citizens who received the thinking treatment were asked to reflect on the vaccine policy enforced by the government, otherwise they were not given an opportunity to think about it. The condition of interest to us was the Nudge+ treatment block, where citizens were automatically defaulted into receiving the vaccine with the option to reflect on this policy. Unlike we had pre-registered, we found that the vaccine, with or without reflection, reduced one's self-reported intentions to accept the vaccine – in other words, we noted a backfire effect. However, we observed that when citizens were additionally made to think about the vaccine policy in the nudge treatment, the thinking washed out all public support for this vaccine, which existed in the standalone nudge treatment condition without any reflection on it. This experiment indicated the nuanced role reflection plays in evaluating vaccination policies as a certain type of public policy that can be disliked by citizens, but it's only when citizens are made to reflect on it that they generate dislike as a signal to the governments.

Similar Nudge+-like vaccination prompts have been also tested by other scholars in the field. In a field experiment with more than 27,000 German participants, Keppeler and co-authors (2022) showed that adding personal possessive pronouns which enables citizens to reflect and take ownership increased vaccination intentions by 2.5 percentage points. In summary, adding the possibility of reflection to a nudge generally improves the effectiveness of the nudge.

## LIMITATIONS AND AUTOCRITIQUE

There are some caveats to Nudge+. First, reflection can backfire. This negative outcome occurs when the goals of the nudge are not aligned with that of the citizens

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or when the nudge is not judged appropriately by its receivers. As discussed earlier, there can be advantages to backfires in ethical terms and in building a long-term relationship with citizens based on trial and error, but it is not clear currently when they happen and to whom.

Second, there seem to be discrepancies between the effectiveness of Nudge+ based on the target behaviours of the nudge. Where decisions are personal and stickier, there can be higher chances of backfire effects. This is clear from evidence on vaccines which are considered to be stickier personal decisions compared to diets. Going forward, tests of Nudge+ also need to ascertain the extent of the persistence of the effect sizes over time and can also speak to the mechanisms of the behaviour change, identifying channels of backfire and psychological reactance.

Third, reflection occurs is not a necessary guard against manipulation. The reflective space may be subject to cues from policymakers that might be hidden from those participating in mini-thinks. There has been a lot of attention to the possibility of framing in the deliberation literature, and efforts to limit through neutral facilitation (Zhong, 2011). Such effects could be more prominent in the short space and time for the mini-thinks, where there is not enough time for facilitation, and of course the think or plus is deliberatively framed through its link to the nudge. Nudge+ may be claiming to bestow autonomy but secretly guiding people to preferred outcomes. Participants may believe they are freely making a choice as there is this opportunity to reflect. In fact, such an apparently open space might cause people to drop their guard in a way they might never do in other situations, such as when reading a glossy pensions brochure when they know for certain what the aims of the promoters are. It resembles the Marxist Herbert Marcuse's term 'repressive tolerance' for liberal freedoms that give the illusion of choice when democracy actually protects dominant economic relationships (Wolff et al., 1969). It behoves designers of Nudge+ to guard against such steering and engage in the highest level of transparency for the plus as well as the nudge.

The final concern is the danger of thinking too much. This concern might sound odd alongside our discussion of the benefits of reflection, but sometimes people find it hard to make decisions when they have too much information to hand, and are overwhelmed by the detail of the choice options that appear in front of them. This overload is captured by Schwartz (2005) in his choice experiments, such as when more choices of type of jam made it harder to come to a decision to buy one jar. The final decision might be made on a whim just to reduce cognitive overload, like grabbing a bag of coffee from the supermarket shelf almost at random because it is hard to make a final choice having seen so many confusing brands and flavours. This problem illustrates the danger of overthinking, which the reflective plus could set off as people might have too much information to handle as they think through what to do.

As discussed above with the collective action problem, thinking or overthinking might cause people to be more aware of their self-interest rather than act in a good collective spirit. Without thinking too much, they agree to the signal to cooperate, such as doing something because others are doing it (social norms here acting to help people realise collective interests). A hypothetical example of selfishness brought about by overthinking is a person who is near to a child about to fall from a railing

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on top of a tall building. The instinctive, unconscious reaction is to reach down with an outstretched arm to try to rescue the child so they can clamber back up. The thinking approach would be to pause, assessing the risk of both the child and the helping person falling to their deaths. As a result, the helper holds back while the hapless child slips off the building falling into the abyss. Which approach is right? It might be the case that relying on instincts for helping might increase risks and generate overconfidence or that evolutionary-determined helping behaviours work to everyone's advantage because they are automatic and use time efficiently rather than delay by thinking. Overall, the risks of overthinking are limited with Nudge+, partly because of the short period of time to think about the problem and the right cues that are in place. Going back to the child once more, it may make sense to think through the risks while the helping hand is reaching out, in that nano-second when there is still a chance to stop, just to check the offer of help is not dangerous in itself. Such time-efficient thinking is very much in the spirit of Nudge+.

## CONCLUSION

There is much potential in deploying new tools to advance the BPP agenda. This chapter has elaborated one example that can foster a bottom-up democratic process alongside BPP. By introducing a think alongside a nudge, Nudge+ delivers agency enhancement whilst not seeking to deny government and individuals the standard battery of nudges that have been tested so thoroughly in recent years. Though the research programme is new, there are promising studies to show the paths for researchers and policymakers to follow. Even backfires (rather than null effects) can be promising (if not too prevalent) as they may show individuals using their agency to decide whether a nudge is good for them or not, which it might not always be. Nudge+ allows citizens to realise their interests when getting nudged. Governments get to learn which nudges citizens prefer. With this interaction between policymakers and citizens, a more collaborative form of governance could emerge (Torfing et al., 2012), wherein bottom-up meets top-down. Needless to say, more research is needed on this and all aspects of Nudge+.

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